

CSA1703 – Artificial Intelligence

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Slot - A

1. 8 PUZZLE PROBLEM

Code:

```
goal = [1,2,3,4,5,6,7,8,0]
print("Enter the initial state (9 numbers):")
state = list(map(int, input().split()))
if state == goal:
    print("Goal State Reached!")
else:
    count = 0
    for i in range(9):
        if state[i] != goal[i]:
            count += 1
    print("Misplaced Tiles =", count)

print("\nInitial State:")
for i in range(0,9,3):
    print(state[i:i+3])

print("\nGoal State:")
for i in range(0,9,3):
    print(goal[i:i+3])
```

Output:

main.py	Run	Output
<pre>1 goal = [1,2,3,4,5,6,7,8,0] 2 3 print("Enter the initial state (9 numbers):") 4 state = list(map(int, input().split())) 5 6 if state == goal: 7 print("Goal State Reached!") 8 else: 9 count = 0 10 for i in range(9): 11 if state[i] != goal[i]: 12 count += 1 13 print("Misplaced Tiles =", count) 14 15 print("\nInitial State:")</pre>		<pre>Enter the initial state (9 numbers): 0 1 7 6 5 3 2 4 8 Misplaced Tiles = 8 Initial State: [0, 1, 7] [6, 5, 3] [2, 4, 8] Goal State: [1, 2, 3] [4, 5, 6] [7, 8, 0] === Code Execution Successful ===</pre>